AI Resume-to-Project Recommendation System

Mini Project Documentation

Developed Using: Base44 AI & Replit

An AI-powered career guidance application that analyzes a student's current skills, identifies skill gaps for a target job role, recommends suitable projects, and generates a personalized learning roadmap.

Project Link: https://replit.com/refer/nangideepthi

1. Project Overview

The **AI Resume-to-Project Recommendation System** is an AI-powered application designed to help students and job seekers identify the skills required for their target job role and improve their career readiness. The user can enter or upload their current skills and specify a target job role. The system analyzes the information, identifies skill gaps, recommends suitable projects, and generates a personalized learning roadmap.

The project was developed using **Base44 AI** and supported through **Replit**.

2. Problem Statement

Students often face difficulty understanding:

- Which skills are required for a particular job role.
- Which skills they are currently missing.
- What projects they should build to improve their resume.
- What learning path they should follow to become job-ready.

This project addresses these challenges by providing an AI-based system that analyzes a student's skills and target role and generates personalized recommendations.

3. Objectives

- Analyze the student's current technical skills.
- Identify the student's target job role.
- Compare existing skills with job requirements.
- Identify important skill gaps.
- Recommend projects based on missing skills.
- Generate a personalized learning roadmap.
- Help students improve their resumes and career readiness.

4. Key Features

Feature	Description
Skill Input	Allows students to enter their current technical skills and experience.
Target Job Role	Allows users to specify their desired career role.
AI Skill Gap Analysis	Analyzes current skills and identifies skills that need improvement.
Project Recommendations	Suggests projects that help users practice and demonstrate missing skills.
Personalized Roadmap	Generates a step-by-step learning and project roadmap.

Feature	Description
Career Guidance	Helps students understand how to become better prepared for their target role.

5. Technologies Used

- Base44 AI – AI-assisted application development.
- Replit – Development, testing, and project prototyping/deployment.
- AI/LLM Concepts – Skill analysis and personalized recommendations.
- Web Technologies – User interface and application functionality.

6. System Workflow

User

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Enter / Upload Skills

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Select Target Job Role

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AI Analyzes Skills

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Compare Skills with Job Requirements

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Identify Skill Gaps

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Recommend Suitable Projects

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Generate Learning Roadmap

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Career Preparation

7. Example

Input

- **Current Skills:** C, Java, Python, Basic HTML/CSS
- **Target Role:** Python Developer

AI Analysis

- **Existing Skills:** Python and programming fundamentals.
- **Skill Gaps:** Advanced Python, SQL, Git/GitHub, APIs, Flask/Django, and database management.

Recommended Projects

- Student Management System
- REST API using Python
- Online Quiz Application
- Personal Portfolio Website
- Database Management Project

Suggested Roadmap

- Improve Python fundamentals.
- Learn SQL and databases.
- Learn Git and GitHub.
- Learn APIs.
- Learn Flask/Django.
- Build real-world projects.
- Update resume and portfolio.
- Apply for internships/jobs.

8. Advantages

- Provides personalized career guidance.
- Helps students identify missing skills.
- Suggests practical projects for skill development.
- Creates a structured learning roadmap.
- Saves time in searching for suitable projects.
- Helps students build stronger resumes and portfolios.

- Supports students preparing for internships and jobs.

9. Future Enhancements

- Resume PDF upload and automatic resume parsing.
- Real-time job-market skill analysis.
- Job recommendations based on user skills.
- Integration with LinkedIn and GitHub profiles.
- Course recommendations from online learning platforms.
- Resume score and improvement suggestions.
- Progress tracking dashboard.
- AI-generated interview questions.
- Personalized interview preparation.
- Skill progress visualization.

10. Conclusion

The **AI Resume-to-Project Recommendation System** provides an intelligent and personalized approach to career preparation. By analyzing a student's current skills and target job role, the system identifies skill gaps, recommends relevant projects, and generates a learning roadmap.

This project demonstrates how AI-assisted development tools such as **Base44 AI** can be used to create practical solutions for student career development and employability.

11. Project Link

Replit: https://replit.com/refer/nangideepthi

12. GitHub Repository Description

AI Resume-to-Project Recommendation System – An AI-powered application that analyzes student skills, identifies skill gaps for their target job role, recommends relevant projects, and generates a personalized career learning roadmap.

13. Keywords

AI • Resume Analysis • Skill Gap Analysis • Project Recommendation • Career Guidance • Student Innovation • Base44 AI • Replit • Roadmap • Job Preparation